Project

Copula Models for Insurance Claims

Financial Engineering

You are asked to estimate three different models for insurance claims from the Danish Fire insurance dataset, which comprises claims (in Danish Krone, DKK) recorded between 1980 and 1990, gathered from the Copenhagen Reinsurance Company, and to backtest the models.

The data are contained in the file “danishmulti.csv”.

1) Estimate the parameters of the mixed distribution model (zero-mixed model), as in Section 4.2.1 of [1].
2) Estimate the parameters of the Comb Bernoulli model in [1], with lognormal marginals and Gaussian copula.
3) Estimate the parameters of the Semi-Parametric Comb Bernoulli model; consider Estimator 2 in [2] for the estimate of correlation.
4) Compute the 95% confidence intervals of the parameters of the three models with the bootstrap technique (cf. Sec.2.4.2 in [1]).
5) Consider the time window from 01/01/1984 to 31/12/1990. For the three models mentioned in points 1) to 3), backtest the 95% and 99% quantiles of total losses, considering:
    a. The models estimated over the time window from 01/01/1980 to 31/12/1983.
    b. The models re-estimated on each value date from 01/01/1984 to 31/12/1990, using a 4-year rolling window.

References

[1] Baviera, R., Manzoni, P., and Massaria, M.D., 2026. Modelling dependence in sparse time series of Insurance Claims, Mimeo.

[2] Baviera, R., Manzoni, P. and Massaria, M.D., 2026. The role of expert judgement in insurance internal model for operational risk, Mimeo.

Realize a library in Matlab.

Room: Massaria

Delivery address: financial.engineering.polimi@gmail.com.